



MPEG-Genomics Terms (For Biologists)

Term	Definition
MPEG-G	ISO standard for compressing genomic data (e.g., FASTQ, BAM, VCF) with efficient storage, selective access, and streaming features.
Genie	A reference software implementation of the MPEG-G standard for compression and decompression of genomic data.
Compression	The process of reducing file size by encoding data more efficiently (e.g., FASTQ → MPEG-G format).
Decompression	Restoring compressed MPEG-G files back to original formats (e.g., MPEG-G → FASTQ).
Selective Access	Ability to retrieve only specific regions, reads, or metadata from a compressed file without decompressing the entire file.
Reference-based Compression	Technique where sequences are stored as differences relative to a reference genome to reduce file size.
Access Unit	A segment of MPEG-G data that can be accessed independently, such as reads for a specific genomic region.
Metadata	Supplementary information in MPEG-G files (e.g., sample name, preparation method, sequencing platform).
Privacy Rules / Encryption	MPEG-G allows encryption of specific genomic regions (e.g., HLA loci), enabling selective access with authorization.
Streaming API	Interface to access compressed genomic data on-the-fly, without full decompression.
Indexing	Feature of MPEG-G that supports fast querying and access to specific reads, samples, or genomic regions.



Docker & Notebook-Based Workflow Terms (For Biologists)

Term	Definition
Docker	A platform to run software in isolated containers, ensuring compatibility across systems. Useful for running Genie without setup hassles.
Docker Container	A preconfigured software environment that can run MPEG-G tools.
Mounting Volumes	Linking folders from your local computer into Docker so the container can access input/output files.
Command-Line Interface (CLI)	A text interface for running Docker and Genie commands like compression or decompression.
Environment Variables	Parameters passed into a Docker container to define paths, settings, etc.
Base Image	The minimal OS image used to build the Docker container for Genie.
Wrapper Script	A helper script (Python or Bash) to automate repetitive Docker tasks like compressing multiple FASTQ files.
Notebook (Jupyter/Python)	A web-based interface where users can run and visualize Python code for genomic compression and downstream ML workflows.
Batch Processing	Running the same process (e.g., compression) over multiple files or samples.